

Behrooz Farkiani

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Fields of Interest

- Networked Systems
- Holistic System Design and Implementation
- Designing Efficient Algorithms for Real-World Problem

Education

2022–2026 Ph.D., Computer Science, Washington University in St. Louis, USA
Supervisor: Prof. Patrick Crowley, Applied Research Laboratory
2012–2014 M.Sc., Information Technology Engineering – Computer Networks,
Amirkabir University of Technology, Tehran, Iran
2007–2012 B.Sc., Information Technology Engineering,
Amirkabir University of Technology, Tehran, Iran

Notable Publications

For a complete list of publications: <https://scholar.google.ca/citations?user=iV7ANb0AAAAJ>

- [1] B. Farkiani, J. DeHart, J. Parwartikar, F. Liu, K. Yang, P. Crowley “The Hermes: A General Purpose Proxy-enabled Networking Architecture,”
- *Languages:* Go, Kotlin, Swift, Lua, Bash, Python
 - *Stack:* Docker, Envoy Proxy, Ansible, PostgreSQL, CouchDB, Influxdb, Prometheus, Grafana, Open Policy Agent, gRPC
 - *Contributions:* A general-purpose overlay architecture designed and implemented to enhance service delivery and communication efficiency.
- [2] **B. Farkiani**, B. Bakhshi, S. A. Mirhassani, T. Wauters, B. Volckaert, and F. D. Turck, "Prioritized Deployment of Dynamic Service Function Chains," IEEE/ACM Transactions on Networking, pp. 1–15, 2021, DOI: 10.1109/TNET.2021.3055074
- *Language:* Java
 - *Stack:* IBM CPLEX, MOSEK solver
 - *Contributions:* A detailed service function chain embedding problem, with an exact algorithm 8x faster than commercial solvers, and a fast near-optimal heuristic
- [3] J. Moeyersons, **B. Farkiani**, B. Bakhshi, S. A. Mirhassani, T. Wauters, B. Volckaert, and F. D. Turck, "Enabling Emergency Flow Prioritization in SDN Networks," in 2019 15th International Conference on Network and Service Management (CNSM), Oct. 2019, pp. 1–8, DOI: 10.23919/CNSM46954.2019.9012696

- *Language:* Python, Java
- *Stack:* IBM CPLEX, MySQL, OpenFlow
- *Contributions:* Flow prioritization, routing, and bandwidth assignment for SDN switches

Notable Technical Experiences

- System Design
 - Designed and prototyped two networking architectures. The first utilized NFV and SDN concepts for efficient service function chain orchestration, using OpenStack, OpenBaton, and OpenDaylight. The second, Hermes, was a novel, complex architecture, built from scratch using six programming languages and a combination of commercial and open-source components.
- Linux
 - Developed a Debian-based update and recovery system with failure detection and image rollback to earlier snapshots, using a modified kernel, initrd, and Btrfs. Also developed a high-speed, non-blocking web server in C using epoll.
- Go
 - Leveraged Go's template library, networking, and concurrency features to design the Hermes overlay manager for proxy placement and configuration updates. Used Go and Ansible to automate overlay component lifecycle management.
- Python
 - Implemented machine learning algorithms in Python for user traffic classification in the Hermes architecture. Additionally, used Python for conducting performance tests and implementing various use cases, taking advantage of its fast prototyping features.
- Java
 - Developed a detailed discrete event network simulator (over 50k lines) in Java to simulate network events, including server migration, traffic management, and routing. Integrated with IBM CPLEX libraries to solve NP-hard problems.

Notable Employments

2022–	Graduate Research Assistant, Washington University in St. Louis, USA
2021–2021	IT Consultant, Self Employed, Tehran, Iran
2015–2020	Graduate Research Assistant, Amirkabir University, Tehran, Iran
2018–2019	Internship on Software Defined Networking, Ghent, Belgium
2018–2020	Head of IT section, The Industrial Engineering and Productivity Research Center, Amirkabir University, Tehran, Iran
2017–2018	IT consultant, Technical and Vocational University, Tehran, Iran
2015–2017	Technical lead of an SDN-NFV networking architecture, Amirkabir University, Tehran, Iran